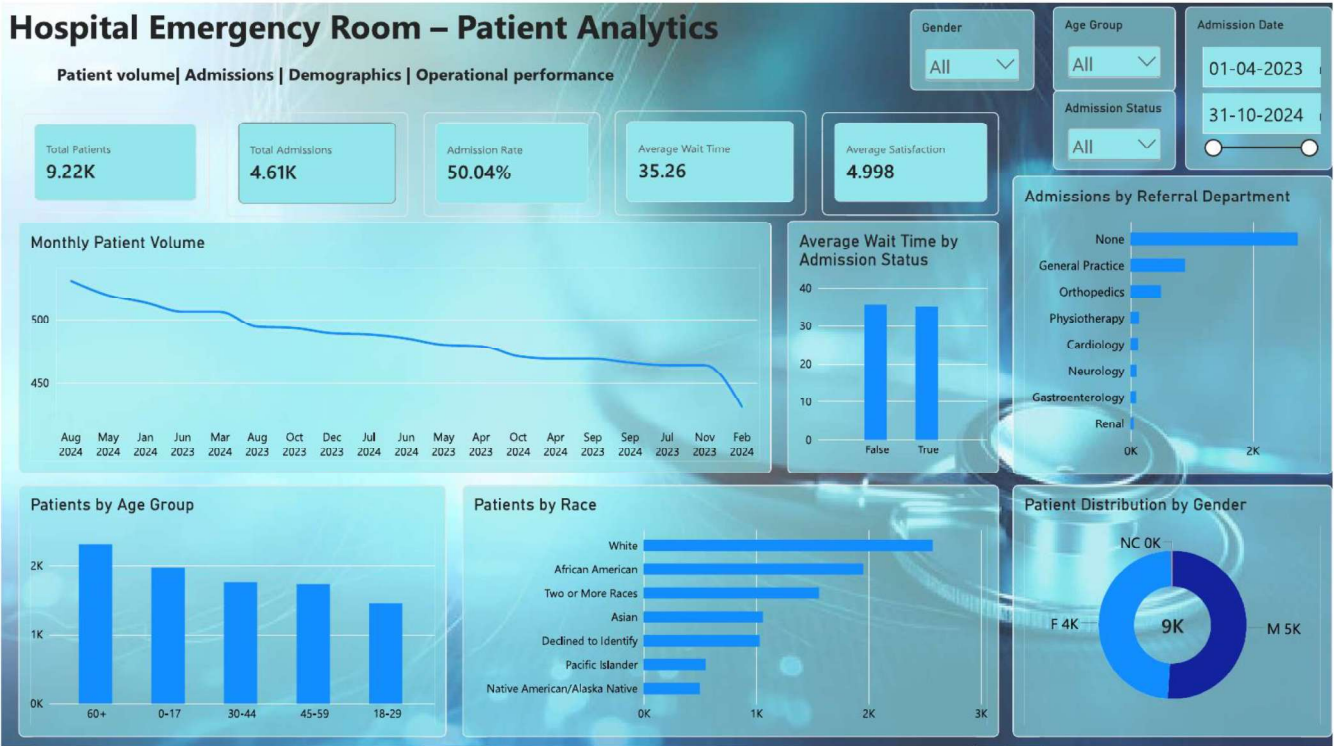


HOSPITAL EMERGENCY ROOM

Patient Analytics Project Report

Patient Volume | Admissions | Demographics | Operational Performance



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Prepared as a Data Analytics Portfolio Project

Tools used: Microsoft Excel/CSV data preparation and Microsoft Power BI

1. Project Overview

This project analyses Hospital Emergency Room (ER) patient data to understand patient volume, admissions, demographics, waiting time and patient satisfaction. The analysis was presented through an interactive Power BI dashboard so that important operational indicators can be reviewed in a simple and business-friendly format.

The report is based on the Hospital ER_Data dataset and the final Hospital Emergency Room – Patient Analytics dashboard. The dashboard covers patient records from April 2023 to October 2024.

2. Business Objective

The main objective of this project is to convert raw emergency room patient data into useful information for management and operational review.

- Understand the overall patient volume and admission level.
- Monitor admission rate and compare admitted and non-admitted patients.
- Analyse patient demographics such as age, gender and race.
- Understand the distribution of patients across referral departments.
- Monitor average patient waiting time.
- Review available patient satisfaction information.
- Provide an interactive dashboard for quick filtering and decision support.

3. Dataset Description

The source dataset contains 9,216 patient records and 12 fields. It includes patient identification information, admission date, demographic information, referral department, admission status, satisfaction score, waiting time and the Patients CM field.

Field / Area	Description
Patient information	Patient ID, first initial and last name
Admission details	Admission date and admission flag
Demographics	Gender, age and race
Referral	Department referral information
Patient experience	Patient satisfaction score and waiting time
Additional field	Patients CM recorded as a binary 0/1 field

Note: The source data does not provide a business definition for the Patients CM field. Therefore, it is not interpreted as a specific clinical or operational measure in this report.

4. Data Preparation and Analysis Approach

The data was reviewed and prepared for analysis before building the dashboard. The main preparation activities were focused on date handling, demographic grouping, referral categorisation and KPI creation.

- Patient admission date was used for time-based analysis.
- Patient ages were grouped into 0–17, 18–29, 30–44, 45–59 and 60+ categories.
- Records without a department referral were presented as 'No Referral' in the dashboard.
- Admission status was used to compare admitted and non-admitted patients.

- Average waiting time and average satisfaction were used as operational and experience indicators.
- A date-based filtering structure was used in Power BI for interactive analysis.

5. Key Performance Indicators

The dashboard provides five primary KPI cards. These indicators give a quick view of the overall emergency room performance represented in the dataset.

KPI	Value	Interpretation
Total Patients	9,216	Total patient records available for analysis.
Total Admissions	4,612	Number of patients marked as admitted.
Admission Rate	50.04%	Approximately half of the patients were admitted.
Average Wait Time	35.26 minutes	Average recorded patient waiting time.
Average Satisfaction	4.998 / 10	Average satisfaction among records with a recorded score.

6. Dashboard Analysis and Key Findings

6.1 Patient Volume

The dashboard contains 9,216 patient records for the period from April 2023 to October 2024. The monthly patient volume visual helps management review changes in patient activity over the reporting period. The dashboard shows variation in monthly volume, with the displayed series ending at a lower volume level than several earlier points.

6.2 Admissions

There are 4,612 admitted patients against 4,604 non-admitted patients. This results in an admission rate of 50.04%, showing a nearly balanced split between the two admission outcomes.

This balance is useful for operational analysis because the dashboard can be used to compare waiting time and other measures between admitted and non-admitted patients.

6.3 Age Group Analysis

Age Group	Patients	Share of Patients
0–17	1,971	21.4%
18–29	1,452	15.8%
30–44	1,755	19.0%
45–59	1,731	18.8%
60+	2,307	25.0%

The 60+ age group is the largest group with 2,307 patients, while the 18–29 group is the smallest with 1,452 patients. This indicates a relatively high representation of older patients in the emergency room dataset.

6.4 Gender Analysis

The dataset contains 4,705 male patients, 4,487 female patients and 24 records marked as NC. The gender distribution is therefore fairly balanced, with a slightly higher number of male patients.